BookDoc

A Hospital Management System

Presentation

Advancements in healthcare technologies enhance patient care, improve efficiency, and create safer, sustainable hospital environments, including telemedicine, AI, smart infrastructure, and eco-friendly practices.

Group Members

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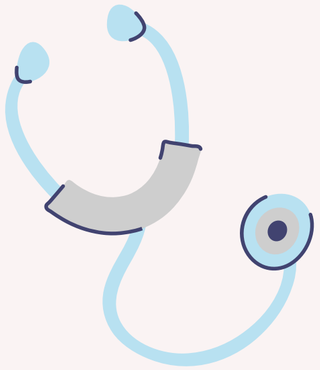




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Introduction



BookDoc is a comprehensive software solution designed to streamline the day-to-day operations of healthcare institutions. This system aims to enhance the efficiency, accuracy, and accessibility of medical and administrative services by automating core hospital functions such as patient registration, appointment scheduling, doctor management, billing, inventory control, and medical record keeping.

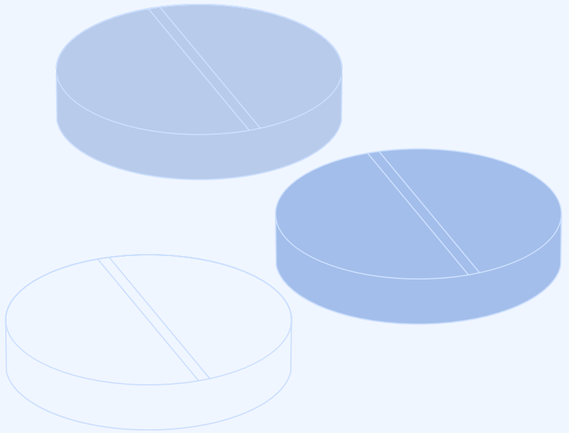


Project Overview





·Digitize patient records for faster access and better data management
·Automate appointment scheduling to reduce patient wait times
·Streamline billing and inventory processes
·Improve hospital efficiency and reduce administrative workload



Stakeholders and Actors



🏥 Hospital Administration

Oversee implementation, monitor system usage, and manage operations.

🩺 Doctors

Access and update patient records, manage treatment history.

👫 Patients

Schedule appointments, view medical records, and access personal health info.

💻 IT Team

Ensure system functionality, perform maintenance, and manage data security



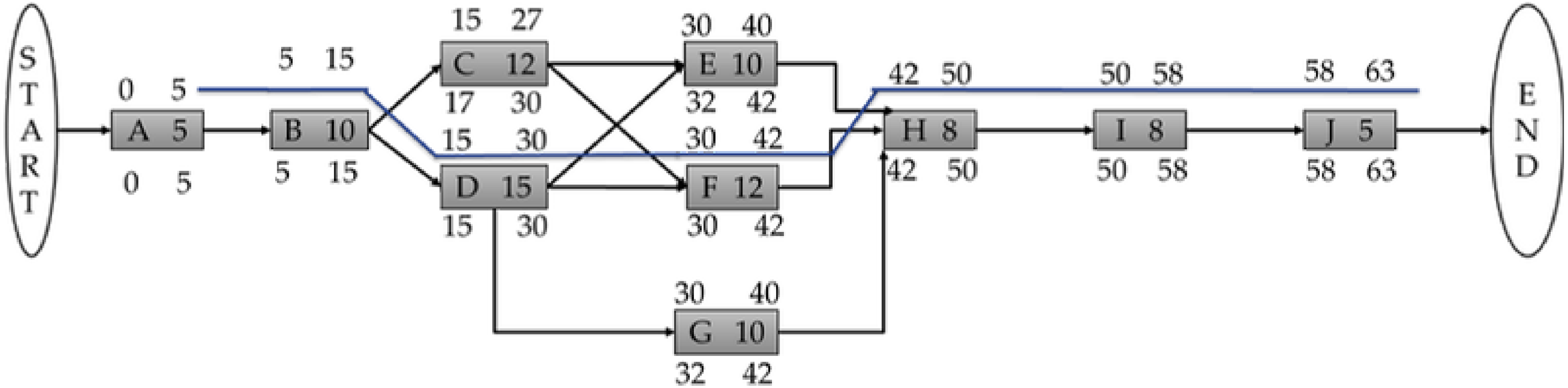
Critical Path Method



Activity ID	Activity Name	Predecessor	Duration
A	Requirement Gathering & Analysis	-	5
B	System & Database Design	A	10
C	User Authentication Module	B	12
D	Patient Management Module	B	15
E	Appointment Scheduling Module	C, D	10
F	Billing & Payment Module	C, D	12
G	Doctor & Staff Management	D	10
H	Report Generation & Notifications	E, F, G	8
I	System Integration & Testing	H	8
J	Deployment	I	5

Diagram





The critical path: A-B-D-F-H-I-J

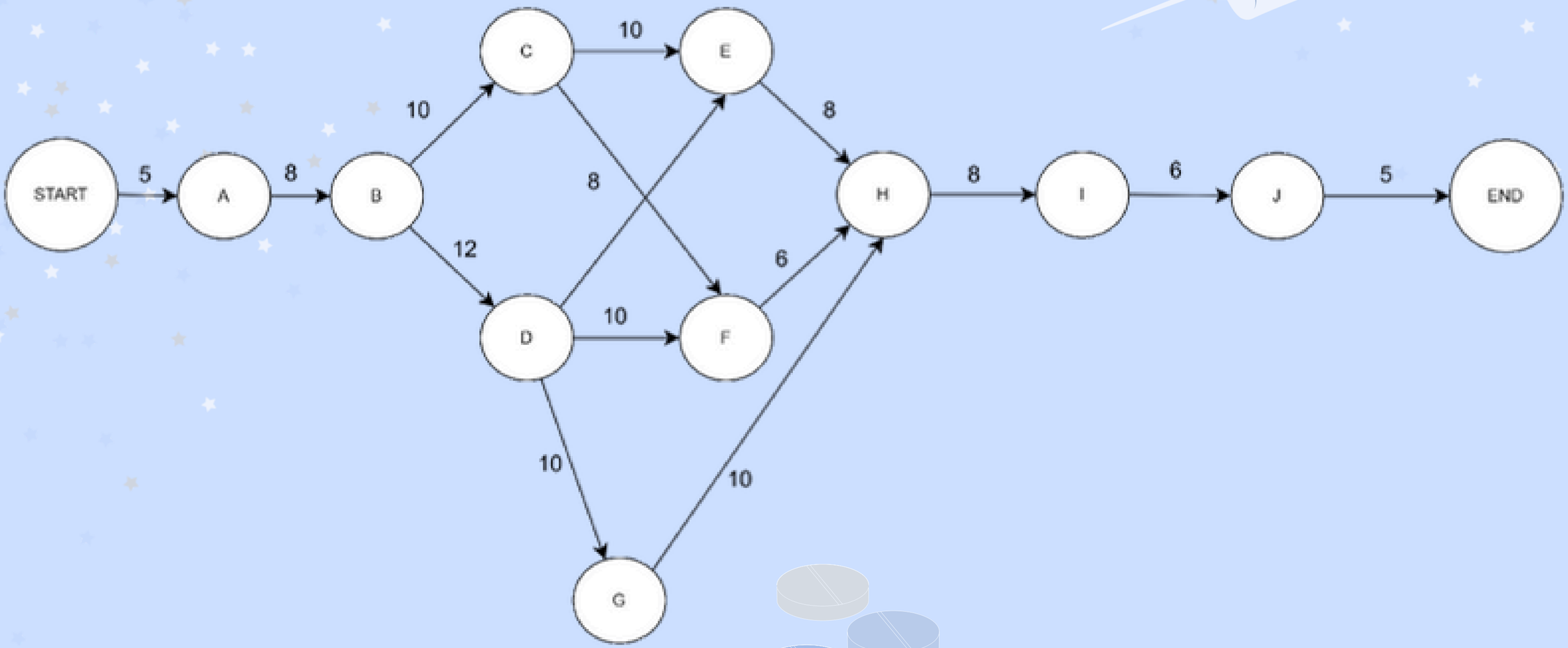
PERT (Program Evaluation & Review Technique)



Activity ID	Activity Name	Predecessor	Optimistic Time (O)	Most Likely Time (M)	Pessimistic Time (P)	TE	**Variance**
A	Requirement Gathering & Analysis	-	6	8	10	8	0.44
B	System & Database Design	A	8	10	12	10	0.44
C	User Authentication Module	B	10	12	14	12	
D	Patient Management Module	B	12	15	18	15	1

E	**Appointment Scheduling Module**	**C,D**	**8**	**10**	**12**	**10**	
F	**Billing & Payment Module**	**C, D**	**10**	**12**	**14**	**12**	
G	**Doctor & Staff Management**	**D**	**8**	**10**	**12**	**10**	**0.44**
H	**Report Generation & Notifications**	**E,F,G**	**6**	**8**	**10**	**8**	**0.44**
I	**System Integration & Testing**	**H**	**8**	**10**	**12**	**10**	**.44**
J	Deployment	I	4	5	6	5	0.11

Diagram



Critical Path : A-B-D-G-H-I-J
Expected Project Length: 64 Days
Project Length Variance: 3.31
σ :1.81

Z= O.8643 =86.43%



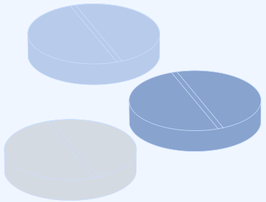
$$z = \frac{T_s - T_e}{\sigma}$$

Where,

T_s - the schedule time to complete the project

T_e - Normal expected project length

σ - Expected standard deviation of the project



Importance of PERT



●Critical Path Analysis

By showing all paths through the network, it helps determine the critical path, which is the sequence of tasks that dictates the minimum project duration. This aids in identifying tasks that cannot afford delays.

●Resource Allocation

By understanding task relationships, project managers can allocate resources more effectively and identify periods of peak demand or slack.

●Improved Scheduling

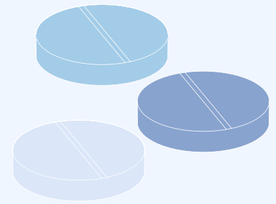
Network diagrams allow project managers to create realistic and efficient schedules by analyzing task dependencies, start times, and deadlines.

●Risk Management

Identifying dependencies and bottlenecks helps in foreseeing potential risks and planning mitigation strategies.

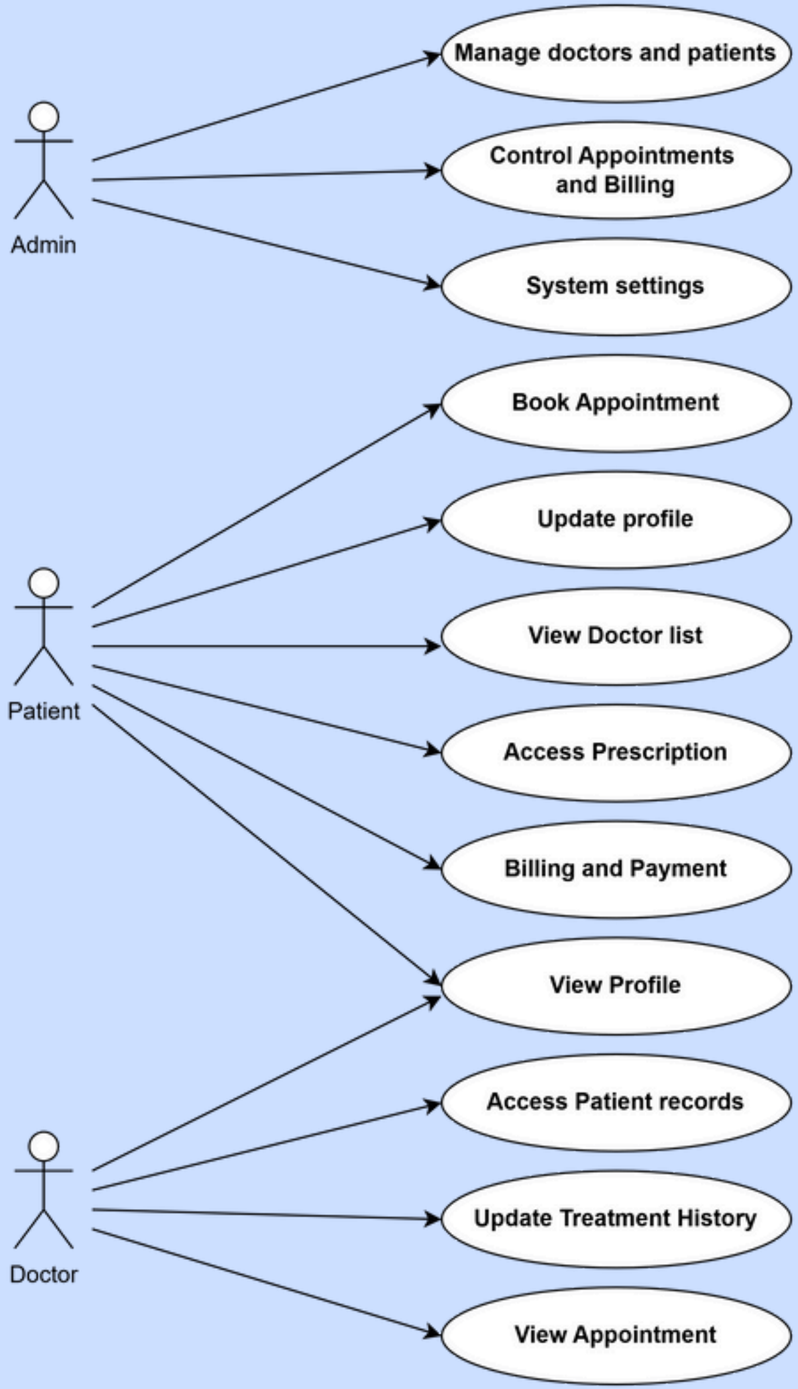
●Progress Monitoring

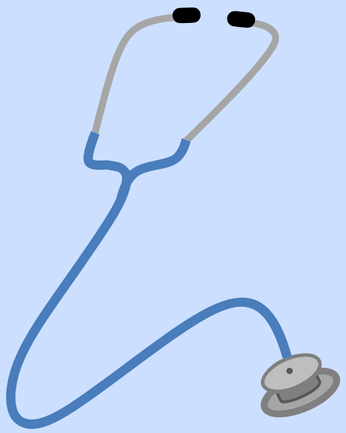
Network diagrams serve as a reference for tracking project progress. They make it easier to identify which tasks are on schedule, ahead, or delayed.



Use Case Diagram



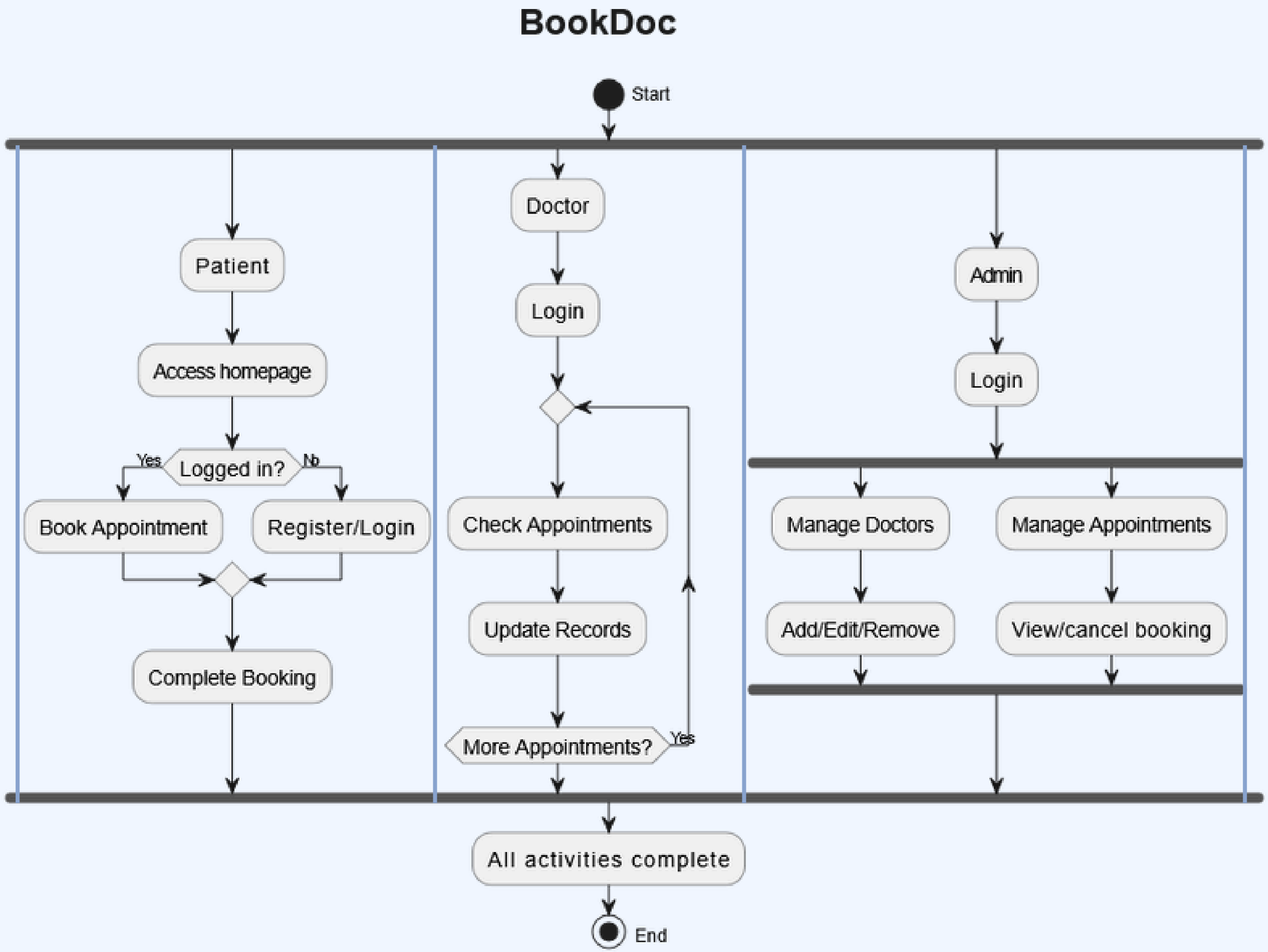

Admin
Manage doctors and patients
Control Appointments and Billing
System settings
Patient
Book Appointment
Update profile
View Doctor list
Access Prescription
Billing and Payment
View Profile
Doctor
Access Patient records
Update Treatment History
View Appointment



Activity Diagram

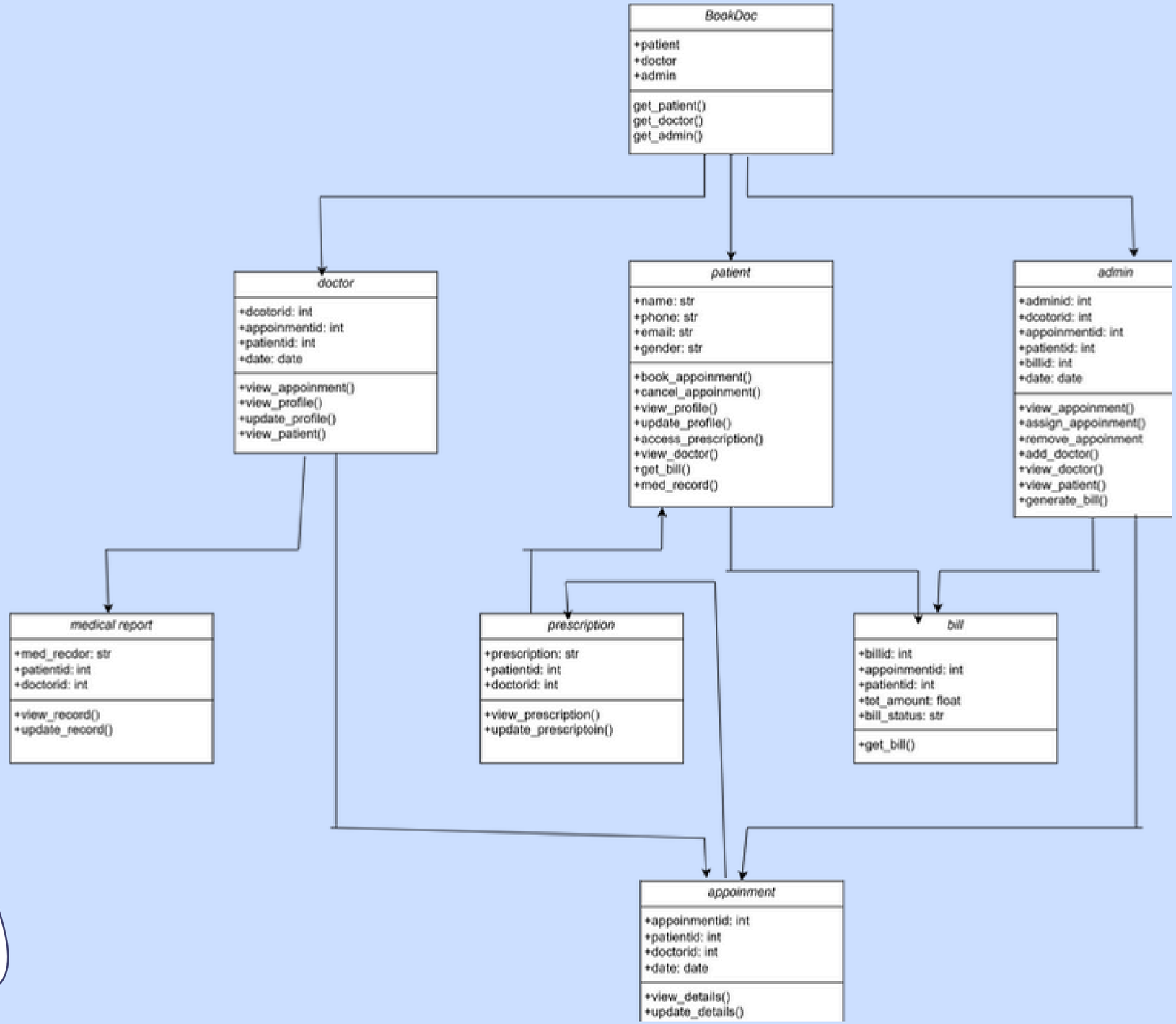


Activity Diagram



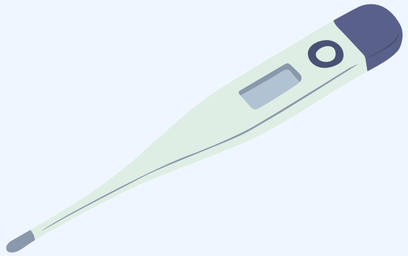
Class Diagramm



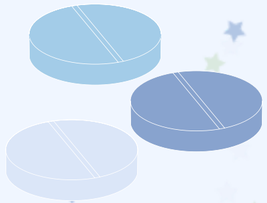

BookDoc
+patient
+doctor
+admin
get_patient()
get_doctor()
get_admin()
doctor
+dcotorid: int
+appoinmentid: int
+patientid: int
+date: date
+view_appoinment()
+view_profile()
+update_profile()
+view_patient()
patient
+name: str
+phone: str
+email: str
+gender: str
+book_appoinment()
+cancel_appoinment()
+view_profile()
+update_profile()
+access_prescription()
+view_doctor()
+get_bill()
+med_record()
admin
+adminid: int
+dcotorid: int
+appoinmentid: int
+patientid: int
+billid: int
+date: date
+view_appoinment()
+assign_appoinment()
+remove_appoinment
+add_doctor()
+view_doctor()
+view_patient()
+generate_bill()
medical report
+med_recdor: str
+patientid: int
+doctorid: int
+view_recoord()
+update_recoord()
prescription
+prescription: str
+patientid: int
+doctorid: int
+view_prescription()
+update_prescriptoin()
bill
+billid: int
+appoinmentid: int
+patientid: int
+tot_amount: float
+bill_status: str
+get_bill()
appoinment
+appoinmentid: int
+patientid: int
+doctorid: int
+date: date
+view_details()
+update_details()

Project Requirements





- Functional Requirements
- Non-Functional Requirements
- Technical Requirements
- Business Requirements





⚙️ Functional Requirements

- Patient registration & profile management
- Doctor & staff management
- Appointment scheduling
- Electronic Medical Records (EMR) management
- Billing & payment processing

⚙️ Non-Functional Requirements

Scalable to support hospital growth
High reliability and system uptime
Secure authentication & data encryption
User-friendly interface for staff & patients

💻 Technical Requirements

Backend: PHP
Database: MySQL
Frontend: HTML, CSS
Hosting: GitHub, LocalHost

📈 Business Requirements

Comply with healthcare standards
Cost-effective hospital management
Enhance operational efficiency





RESOURCE MANAGEMENT

Development Tools: Visual Studio Code

Hosting: GitHub, LocalHost

Database: MySQL(XAMPP)



RISK MANAGEMENT





Data Security Breach

🔒 **Mitigation: Implement encryption, strong authentication, and access controls.**

System Downtime

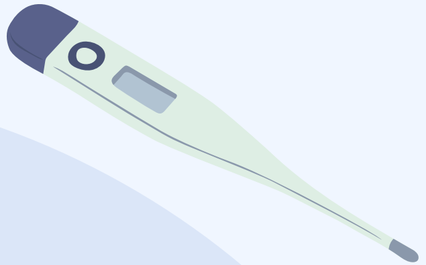
⚙️ **Mitigation: Schedule regular maintenance and system backups.**

Low User Adoption

👩‍🏫 **Mitigation: Conduct training sessions and provide ongoing support.**

Quality Management





✓ Continuous testing & user feedback

✓ Performance, satisfaction & compliance checks

✓ Regular audits & bug tracking

Communication Plan



✓ Continuous testing & user feedback

✓ Performance, satisfaction & compliance checks

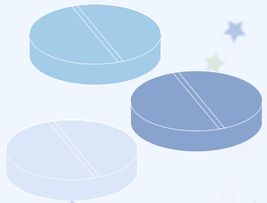
✓ Regular audits & bug tracking



✓ Weekly meetings & email updates

✓ Reports & stakeholder updates

✓ Documentation & system architecture



Analysis of Existing Projects



Doctorola.com: Good for doctor search and appointments; lacks EHR, billing, and hospital system integration.

OpenMRS: Feature-rich and open-source; supports EHR and billing but needs technical expertise.

Epic Systems: Comprehensive and highly integrated; best for large hospitals but expensive and complex.

Surokkha: Limited to vaccine appointments; not suitable for general hospital management.





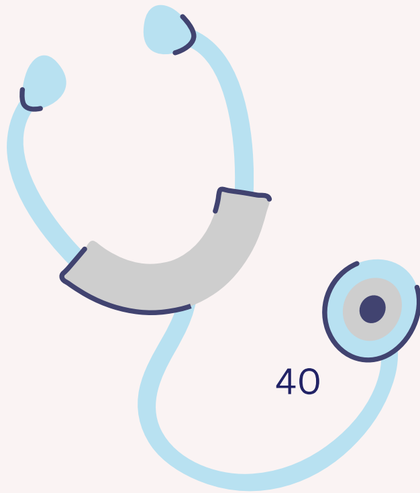


COMPARATIVE ANALYSIS



- Local Systems: Limited features, basic tech, low integration.
- International Systems: Comprehensive, advanced tech, high integration.

Insight: Local platforms are simple but limited; global systems are powerful but complex—there's a need for a balanced, locally-adapted HMS.



GAP ANALYSIS & IMPROVEMENTS



Gaps in Local Systems

- No EHR
- Poor coordination
- Outdated technology

Proposed Solutions

- Real-time schedule sync
- AI-based appointment system
- Mobile app for patients and doctors

Key Features of BookDoc

- Patient & Doctor Dashboards
- Admin Control Panel
- Cloud Backup & AI Engine
- User-friendly, responsive design (PHP, MySQL, Bootstrap)





Implementation Plan



Steps to Incorporate Solutions

- Design database schema
- Develop web app
- Create REST API for mobile integration
- Implement AI module for schedule optimization
- Deploy on cloud

Tools And Technologies

- MySQL/PostgreSQL
- React + Node.js/Django
- Python
- AWS, Firebase



SYSTEM MODULES – USER ACCESS

Homepage Overview

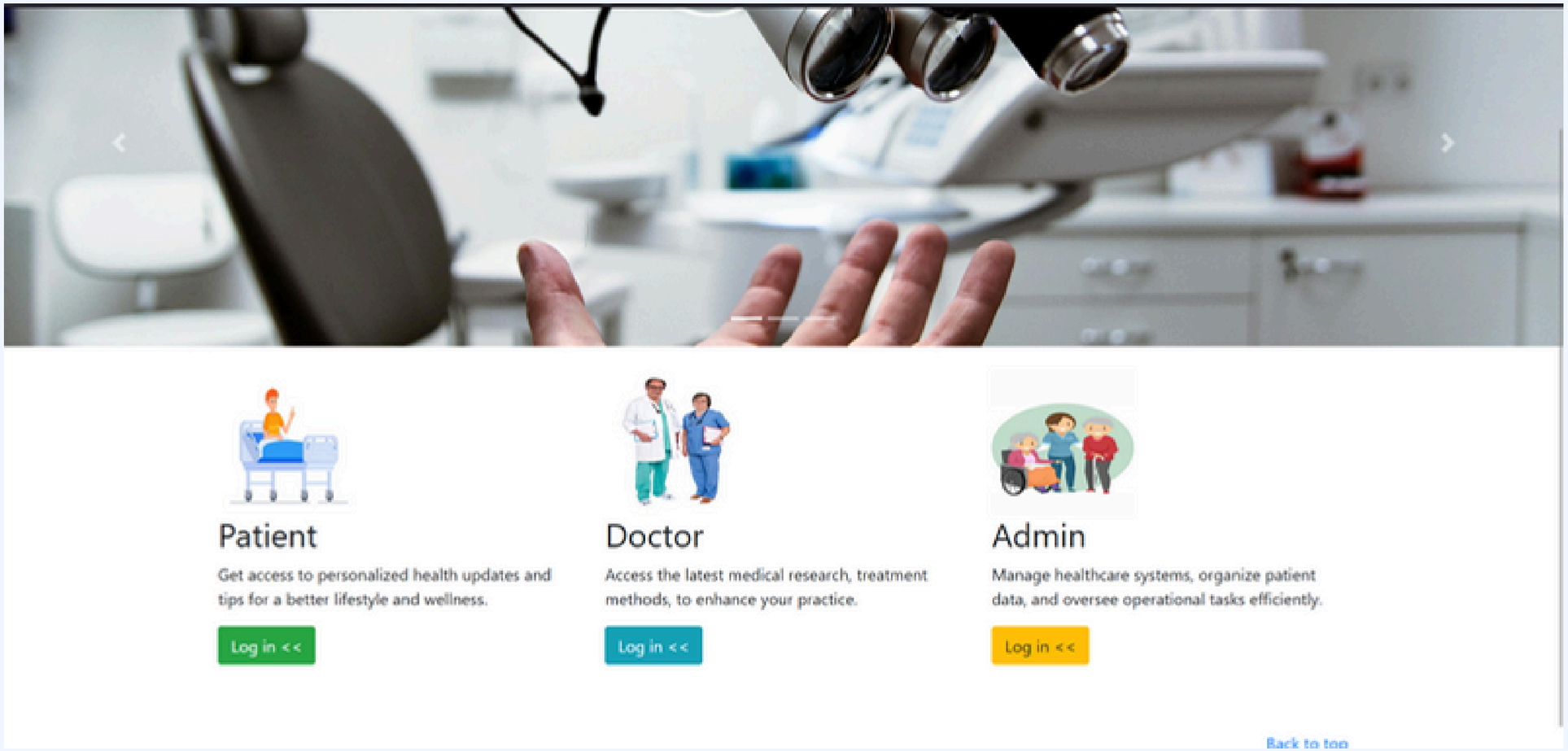
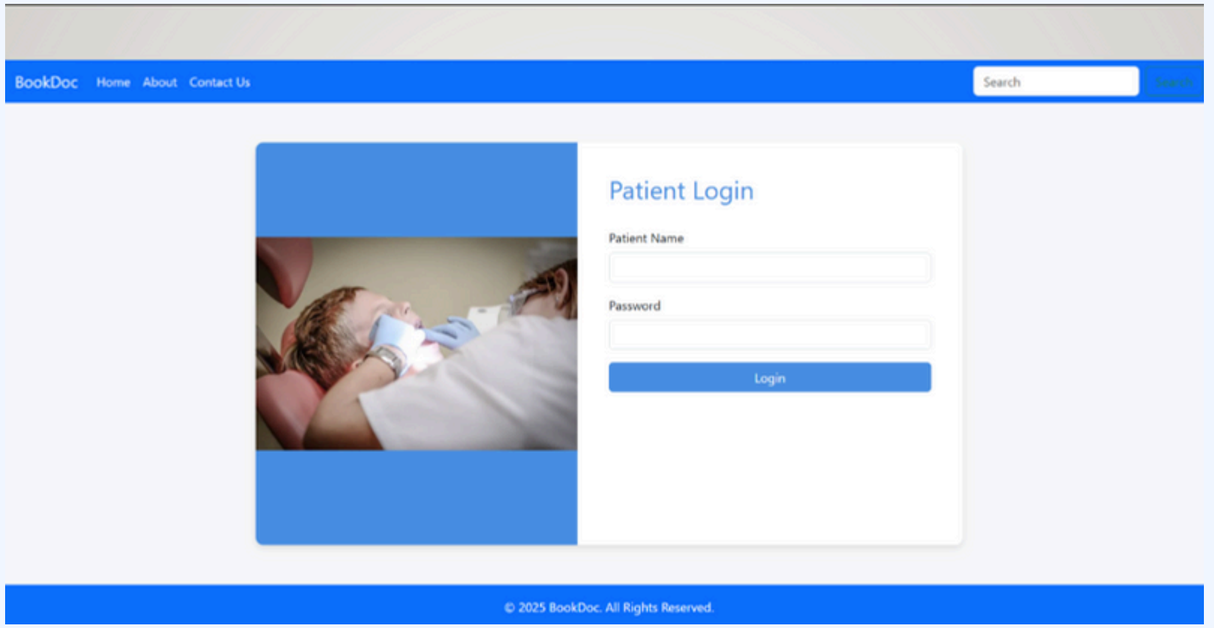
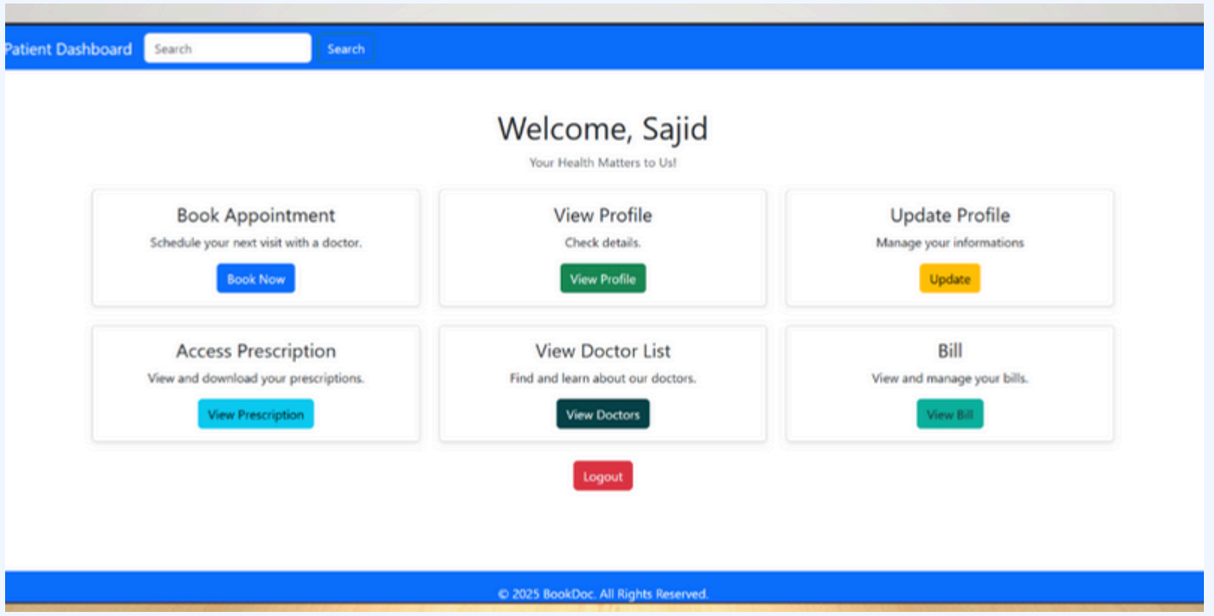


Figure: HomePage

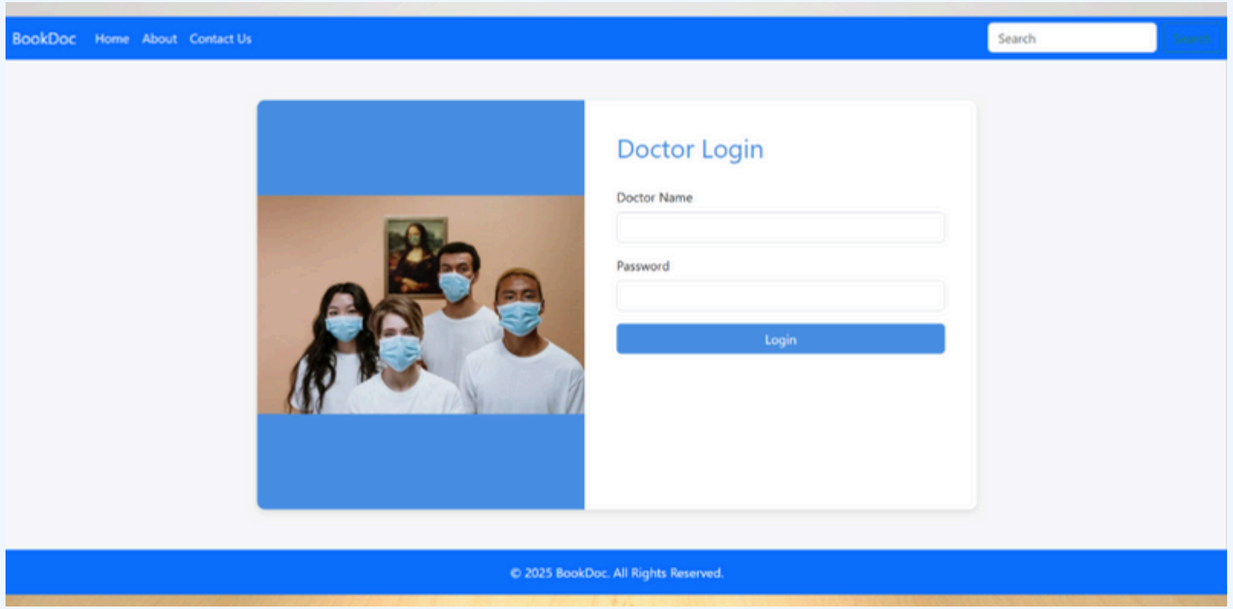
Patient	- View profile - Book Appointment - Update profile - View Doctor list -Access Prescription -Billing & Payment



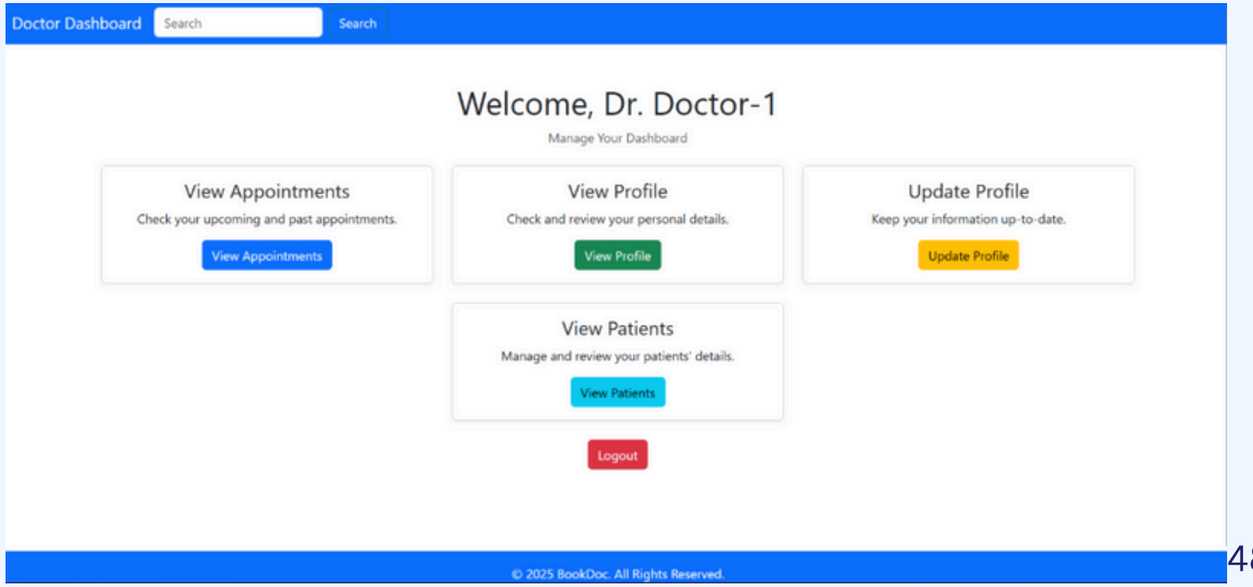




Doctor	- View profile - Access Patient Records - Update Treatment History - View Appointments

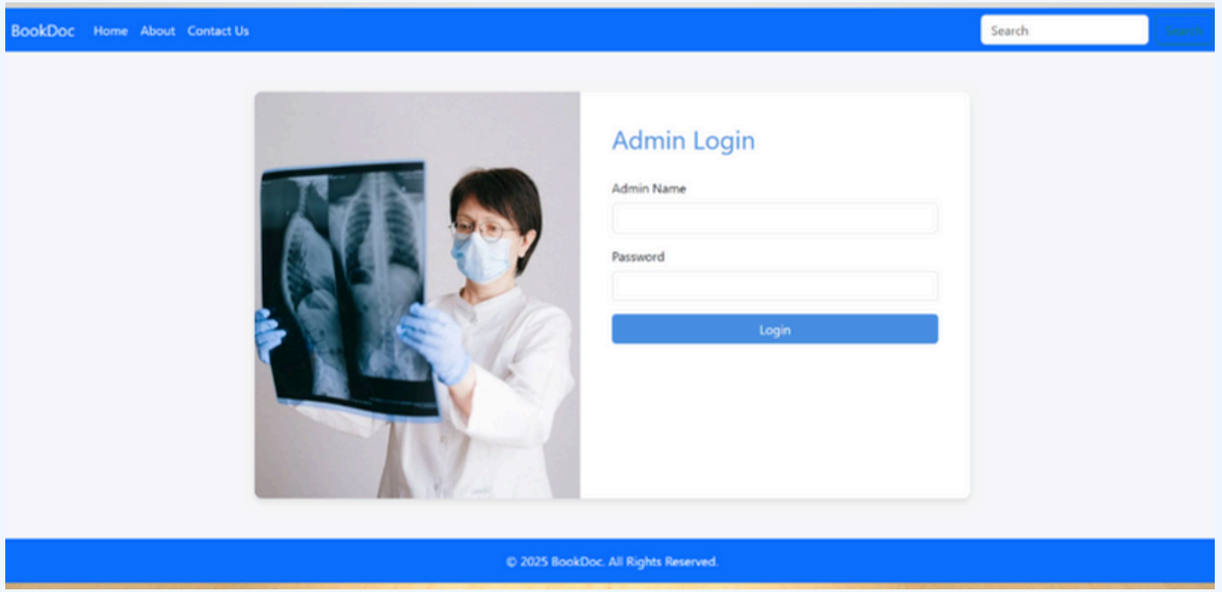


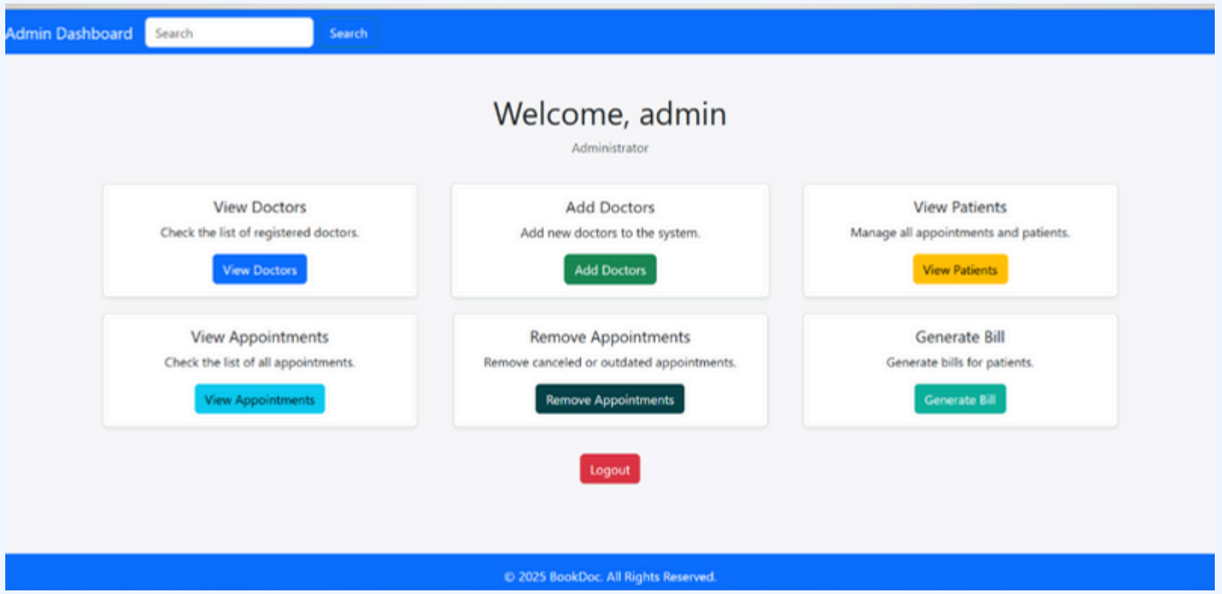




Admin	- Manage Doctors & Patients - Control Appointments & Billing - System Settings







Conclusion





In conclusion, an Online Booking and Appointment System streamlines the booking process for customers and simplifies management for service providers, improving efficiency, reducing errors, and enhancing the overall user experience. By offering real-time availability, secure payments, and automated reminders, it saves time and ensures smoother operations for both users and administrators.





Thank you for your attention

